BPUT HACKATHON 2026

Sustainable Facility and Estate Intelligence Platform for India

From Data to Decisions for Smarter, Sustainable Campuses



PS CODE	: PS-04
TEAM NAME	: TEAM ELITE
TEAM ID	: BH26PS04T039



From Facility Data to Intelligent Decisions

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One platform that answers the questions facility teams ask every day — before the bill arrives.

01
THE PROBLEM
What is going wrong?

02
WHO CARES
Who feels the pain?

03
THE SOLUTION
What are we building?

04
HOW IT WORKS
Architecture & demo

05
WHY US
Research gaps — our edge

06
BUSINESS
Can it make money?

07
IMPACT
Why it matters

08
WHAT'S NEXT
Roadmap

The Problem

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Facilities Generate Data. Decisions Still Happen Manually.

Situation

Current Reality

Water Data
Separate meters, different formats

Energy Data
Multiple meters, buildings, transformers

Maintenance
Mostly manual, reactive

Analysis
Done manually in spreadsheets

Decision
Human dependent and delayed

Problem





The problem is not only resource consumption. The problem is **fragmented visibility** and **delayed decision-making.**



WHO FACES THE PROBLEM?

Built for the People Who Run Facilities

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COLLEGES
GRIHA mandates, tight budgets

HOSPITALS
24/7 energy, patient safety

CORPORATE CAMPUSES
ESG reporting pressure

PSUS & GOVT OFFICES
Mandatory energy audits

INDUSTRIAL ESTATES
Resource-heavy operations

MUNICIPAL FACILITIES
Smart city programs

THEIR DAY SOUNDS LIKE THIS

FACILITY ADMINISTRATOR "TOO MUCH DATA, TOO LITTLE VISIBILITY."

OPERATIONS TEAM "WHICH ISSUE NEEDS ATTENTION FIRST?"

SUSTAINABILITY OFFICER "ARE WE ACTUALLY IMPROVING?"

MANAGEMENT "WHAT HAPPENS IF WE CHANGE THIS?"



RESEARCH PAPER STUDY

RESEARCH GAPS → OUR ENGINEERING ADVANTAGES

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P1 — HIMEUR ET AL. 2021 (APPLIED ENERGY)

OBJECTIVE

TAXONOMY OF AI ANOMALY-DETECTION METHODS; TRENDS & PERSPECTIVES

METHODOLOGY

TAXONOMY OVER ML MODEL, FEATURES, DETECTION LEVEL, PLATFORM, APPLICATION, PRIVACY

LIMITATIONS

NO UNIFIED METRICS; NO REPRODUCIBILITY PLATFORM; ANOMALIES ILL-DEFINED; LABELED DATA ABSENT; PRIVACY RARELY ADDRESSED

RESEARCH GAPS

(I) PRECISE ANOMALY DEFINITION
(II) ANNOTATED DATASETS
(III) UNIFIED METRICS
(IV) REPRODUCIBILITY PLATFORMS
(V) PRIVACY PRESERVATION

P2 — BHARATH ET AL. 2026 (DISCOVER SUSTAINABILITY)

OBJECTIVE

EXPLAINABLE MULTI-ZONE DIGITAL TWIN WITH QUANTIFIED DECISION IMPACT

METHODOLOGY

XGBOOST REGRESSION + SHAP TREEEXPLAINER + SCENARIO SIMULATOR + PAIRED WITH/WITHOUT-XAI TRIALS (N=42)

LIMITATIONS

ALGORITHM-ONLY TRIALS (NO HUMANS); SINGLE BUILDING/CLIMATE; CVRMSE 45.7% > ASHRAE 30%; XAI POLICY 24% FAILURE; ROLE UI UNTESTED

RESEARCH GAPS

HUMAN DECISION-EFFECTIVENESS UNTESTED; CROSS-ZONE COUPLING LIMITED; XAI CONFIDENCE-THRESHOLDING NEEDED

P3 — MUNIR ET AL. 2026 (IET SMART GRID)

OBJECTIVE

SURVEY OF AI/DL/ML FORECASTING METHODS FOR BUILDING ENERGY

METHODOLOGY

COMPARATIVE ANALYSIS OF PHYSICAL, DATA-DRIVEN, HYBRID FORECASTING

LIMITATIONS

REVIEW-LEVEL ONLY; INCONSISTENT DATA FORMATS/HORIZONS; TECHNOLOGY-ORIENTED RATHER THAN PROBLEM-DRIVEN RESEARCH

RESEARCH GAPS

UNSTRUCTURED/NON-STANDARD DATA; 128+ ALGORITHMS, NO DOMINANT ONE; BENCHMARK STANDARDS NEEDED

PROPOSED SOLUTION

MORE THAN A DASHBOARD

EstateIQ

01

AI Anomaly Detection

Spots unusual energy, water behaviour — after checking occupancy, weather and time of day.

02

Short-Term Forecasting

Predicts the next 24 hours to 7 days of demand, so teams act before problems arrive.

03

Explainable AI

Every alert says WHY. SHAP shows which factor drove it — no black boxes.

04

What-If Simulation

“What if we cut HVAC 10%?” Test decisions safely, see estimated impact first.

ELITE ALGORITHM

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01 | TELEMETRY INPUTS

- Building Meters
- Occupancy (PPL)
- Ambient Temp (°C)
- HVAC Schedules

02 | QUALITY & SCREENING

STAGE 1: DATA QUALITY GATE

- Completeness
- Freshness
- Sensor Reliability

STAGE 2: CONTEXT FILTER

EXECUTION PATHS

- Path A: Fast
- Path B: Intelligence
- Path C: Critical

03 | FORECASTING & ANOMALIES

STAGE 3: ADAPTIVE ENSEMBLE

Champion-Challenger

Elite Algorithm

Adaptive Forecasting

STAGE 4: ANOMALY CONSENSUS

- Multi-Signal
- Anomaly Score

04 | EVIDENCE TRUST & IMPACT

STAGE 5: CONFIDENCE ENGINE

Trustworthiness Score

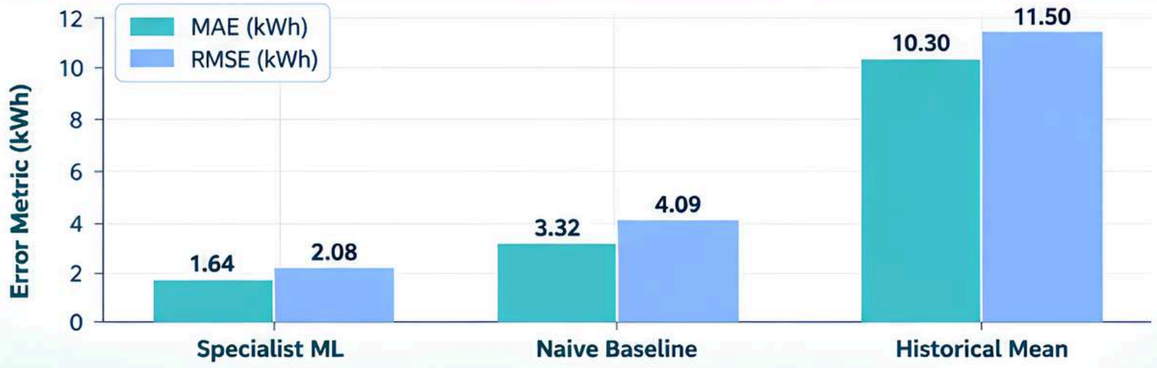
- Data Quality
- Model Agreement
- History
- Competence
- Reliability

Reliable & Actionable Insights

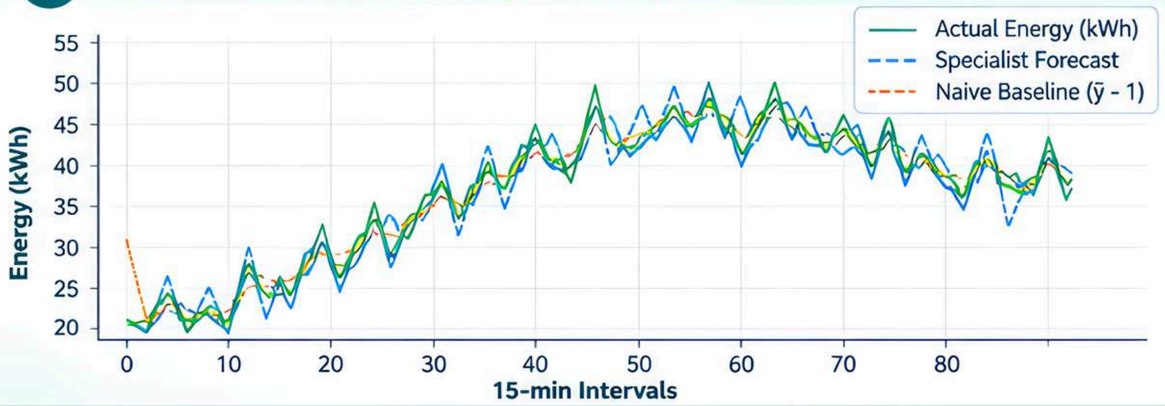
ALGORITHM METRICS

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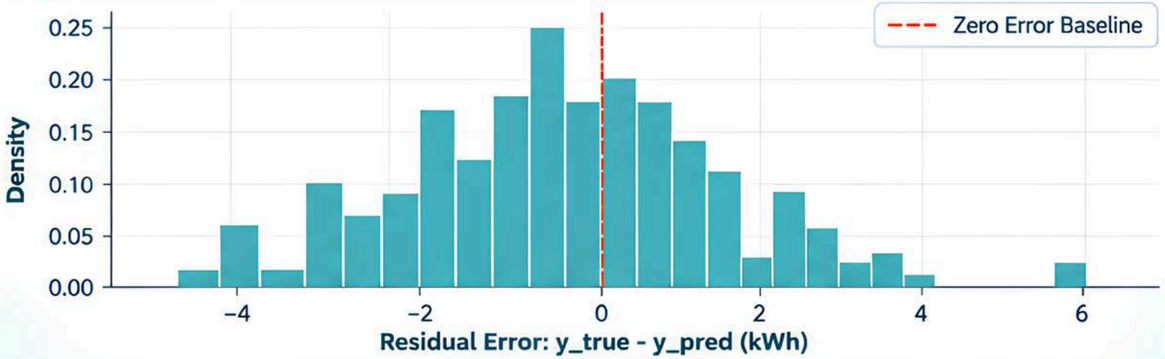
A Model Performance Comparison (MAE vs RMSE)



B Actual vs Predicted Energy Consumption



C Residual Error Distribution (Mean: -0.40, Std: 2.04)



D ROC & Precision-Recall Anomaly Classifier Curves



ELITE ALGORITHM — COMPUTATIONAL COMPLEXITY

Decision-layer Time Complexity

$$O(M + K \log K + P)$$

Combines processing of specialist model signals, action ranking and contextual features.

Auxiliary Space Complexity

$$O(M + K + P)$$

Stores model outputs, candidate actions and contextual features.

Where:

- M = number of specialist-model signals (e.g., forecast, anomaly scores)
- K = candidate actions / recommendations being ranked
- P = contextual features used in fusion/scoring (e.g., asset type, operational context, cost factors)

Elite Algorithm Decision Pipeline

Validate Data → Fuse Signals → Score Impact → Rank Decisions → Recommend Actions

⚠ Illustrative upper bound under these assumptions; excludes model training inference and depends on implementation.

Evaluation values must be validated on the current test dataset.

Tech Stack

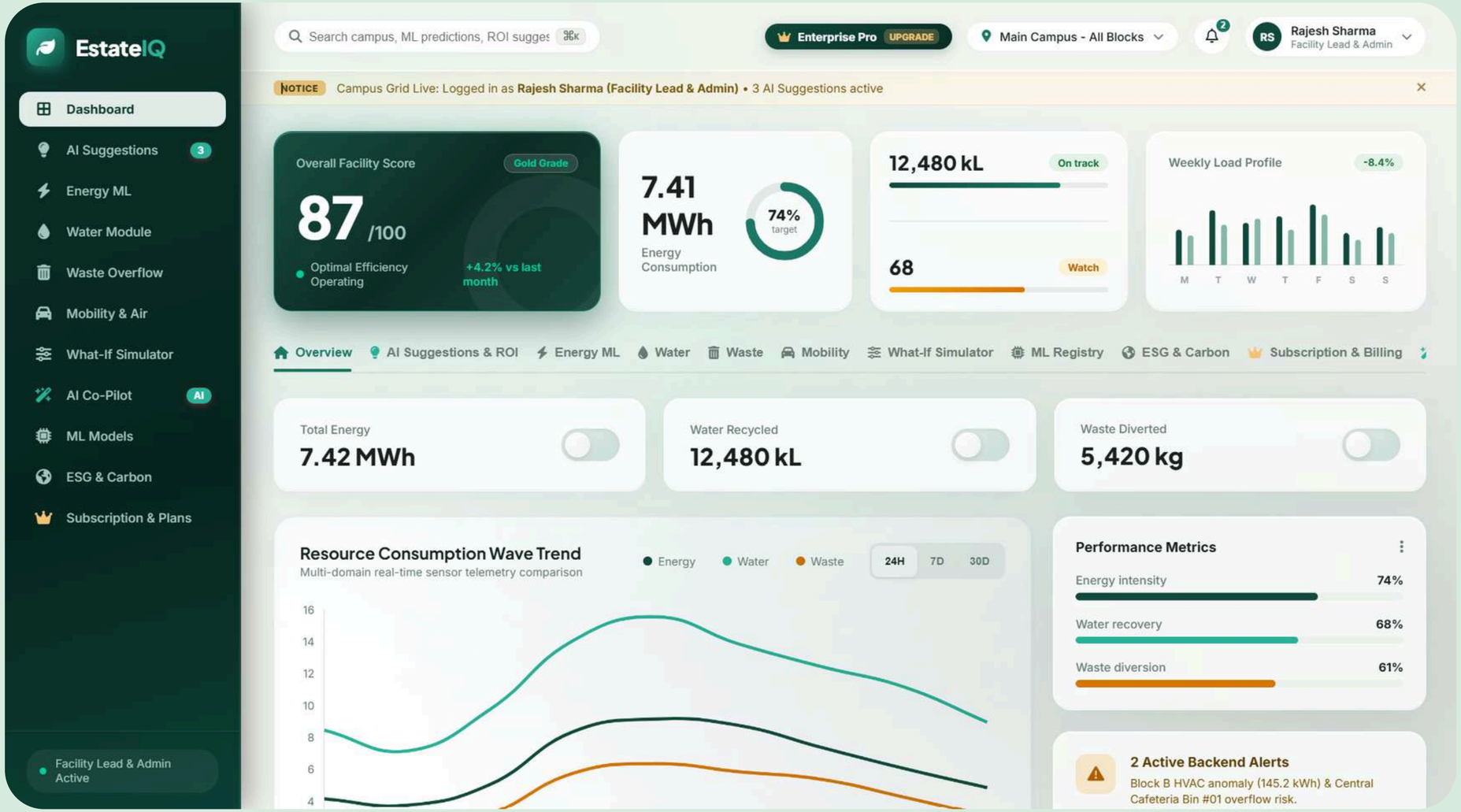
Built to Be Practical, Not Just Impressive

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Category	Technologies
Core & Backend	Python 3.10+, FastAPI, Uvicorn, Pydantic
AI / ML	Scikit-learn, XGBoost, LightGBM, CatBoost, Prophet
Intelligence	SHAP, Anomaly Detection, Forecasting, Optimization
Data & Visualization	Pandas, NumPy, Plotly, GeoPandas, Folium
Frontend	Streamlit
IoT	**Electricity:** ESP-32 Dev Module, PZEM-004T 10A AC **Water :** ESP-32, ZJ-S201 Water Flow Sensor, Ultrasonic Obstacle Sensor **Waste** : ESP-32, HC-SR04 Ultrasonic Sensor **Air** : ESP-32, MQ135, DHT11 Temperature And Humidity Sensor
Security	Advanced Encryption Standard, JWT, Role-Based Authentication

PRODUCT DEMO

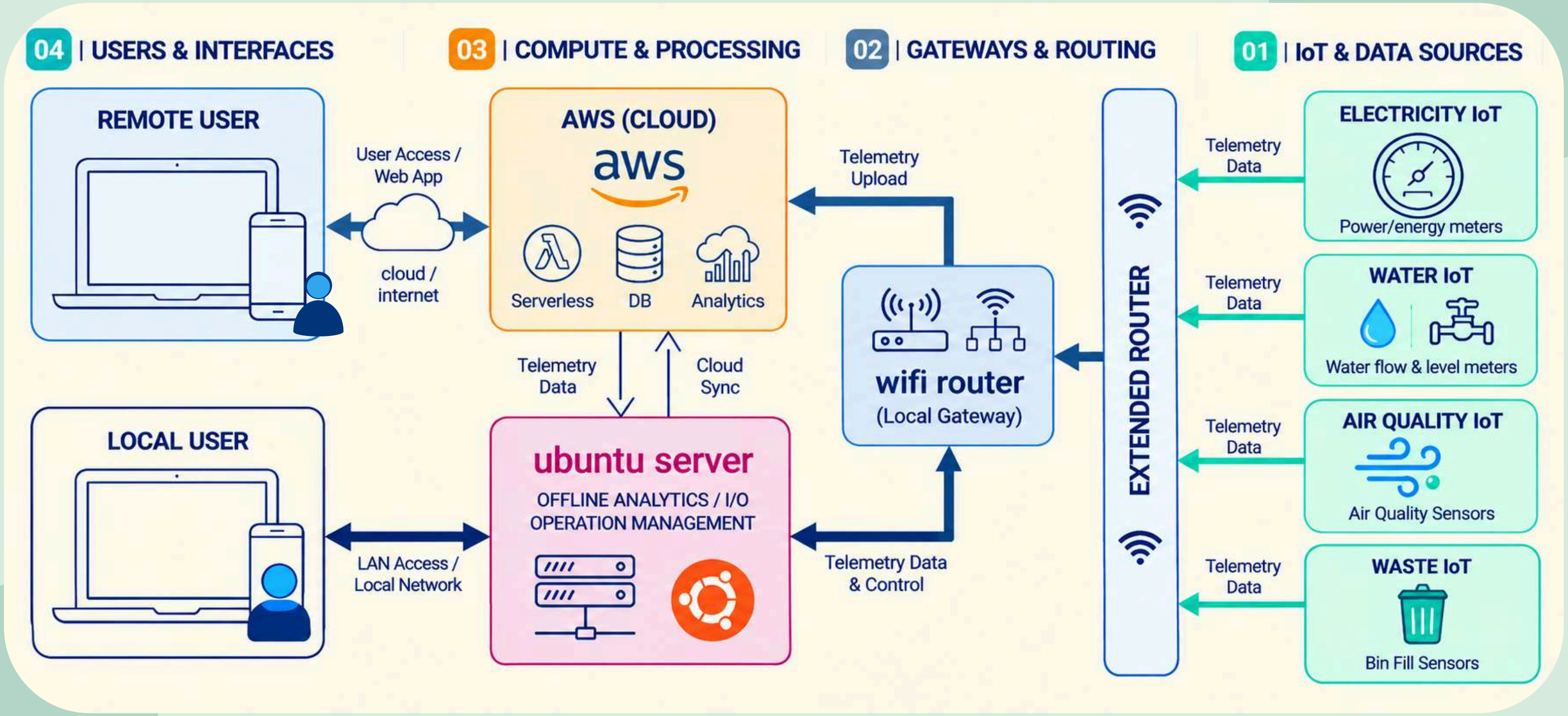
From Alert to Action



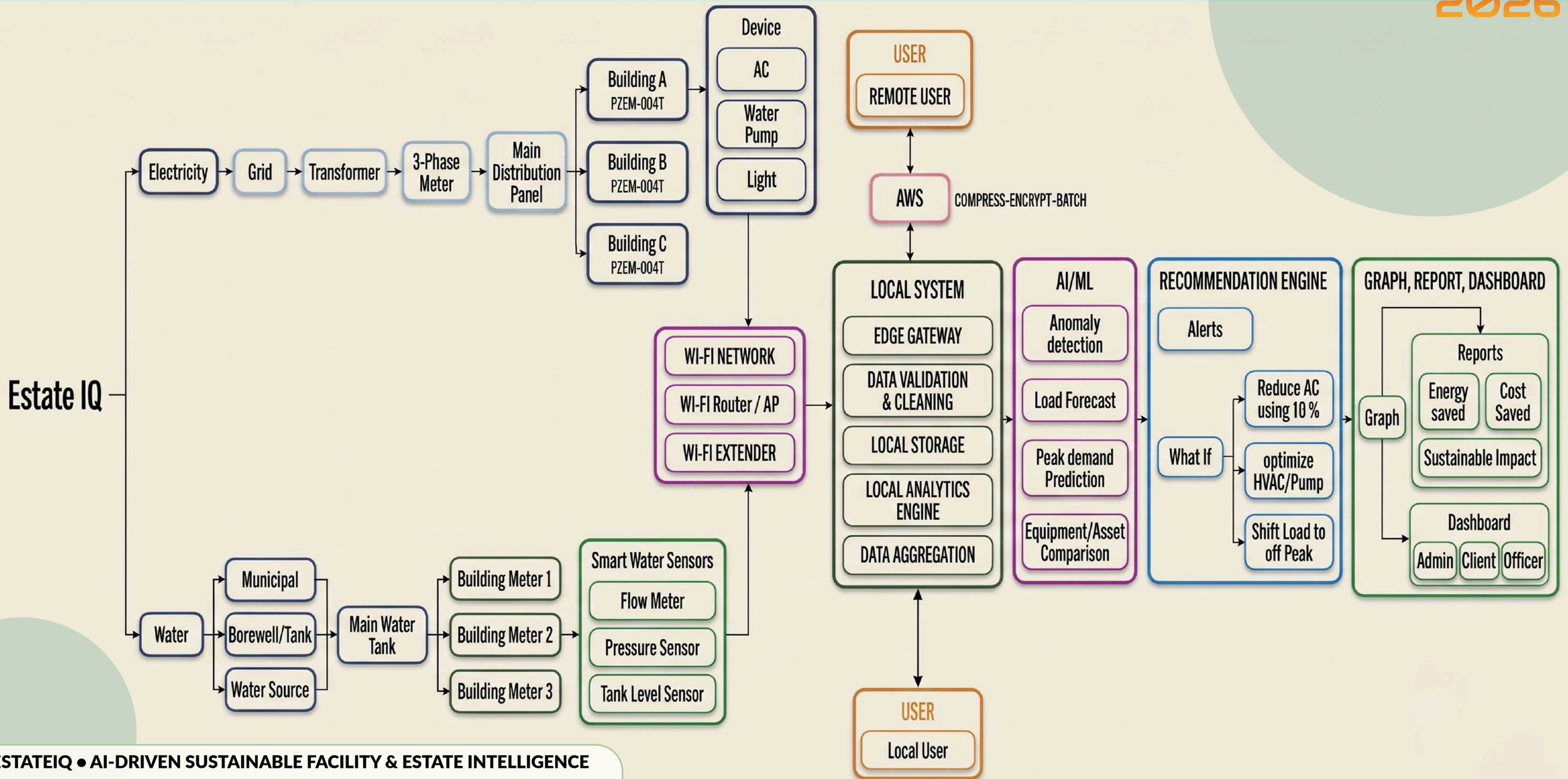
WHAT THE ADMIN JUST DID

1. **SELECTS FACILITY**
   ONE DROPDOWN — COLLEGE, HOSPITAL, PSU...
2. **SEES THE ALERT**
   SEVERITY-RANKED, NOT A WALL OF CHARTS
3. **READS THE WHY**
   SHAP NAMES THE DRIVER: HVAC 47%
4. **CHECKS THE FUTURE**
   24H FORECAST WITH CONFIDENCE BAND
5. **TESTS A FIX**
   WHAT-IF: −9% ENERGY, COMFORT INTACT
6. **DECIDES**
   ACCEPT → ACTION LOGGED IN AUDIT TRAIL

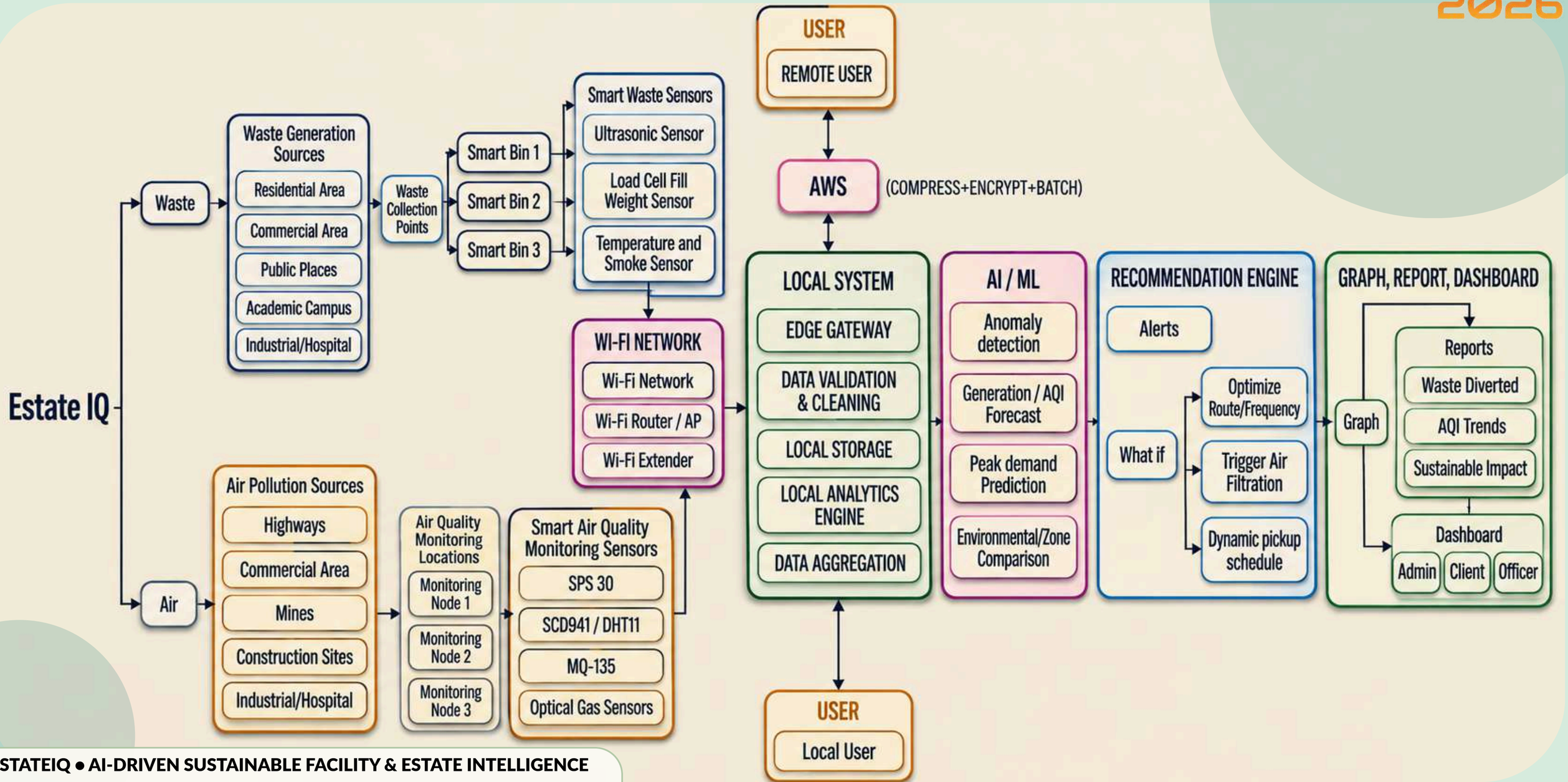
ARCHITECTURE DIAGRAM



Flow Chart


Estate IQ
Electricity
Grid
Transformer
3-Phase Meter
Main Distribution Panel
Building A PZEM-004T
Building B PZEM-004T
Building C PZEM-004T
Device
AC
Water Pump
Light
USER
REMOTE USER
AWS
COMPRESS-ENCRYPT-BATCH
WI-FI NETWORK
WI-FI Router / AP
WI-FI EXTENDER
LOCAL SYSTEM
EDGE GATEWAY
DATA VALIDATION & CLEANING
LOCAL STORAGE
LOCAL ANALYTICS ENGINE
DATA AGGREGATION
AI/ML
Anomaly detection
Load Forecast
Peak demand Prediction
Equipment/Asset Comparison
RECOMMENDATION ENGINE
Alerts
What If
Reduce AC using 10 %
optimize HVAC/Pump
Shift Load to off Peak
GRAPH, REPORT, DASHBOARD
Graph
Reports
Energy saved
Cost Saved
Sustainable Impact
Dashboard
Admin
Client
Officer
Water
Municipal
Borewell/Tank
Water Source
Main Water Tank
Building Meter 1
Building Meter 2
Building Meter 3
Smart Water Sensors
Flow Meter
Pressure Sensor
Tank Level Sensor
USER
Local User

Flow Chart


Estate IQ
Waste
Waste Generation Sources
Residential Area
Commercial Area
Public Places
Academic Campus
Industrial/Hospital
Waste Collection Points
Smart Bin 1
Smart Bin 2
Smart Bin 3
Smart Waste Sensors
Ultrasonic Sensor
Load Cell Fill Weight Sensor
Temperature and Smoke Sensor
WI-FI NETWORK
Wi-Fi Network
Wi-Fi Router / AP
Wi-Fi Extender
Air
Air Pollution Sources
Highways
Commercial Area
Mines
Construction Sites
Industrial/Hospital
Air Quality Monitoring Locations
Monitoring Node 1
Monitoring Node 2
Monitoring Node 3
Smart Air Quality Monitoring Sensors
SPS 30
SCD941 / DHT11
MQ-135
Optical Gas Sensors
USER
REMOTE USER
AWS
(COMPRESS+ENCRYPT+BATCH)
LOCAL SYSTEM
EDGE GATEWAY
DATA VALIDATION & CLEANING
LOCAL STORAGE
LOCAL ANALYTICS ENGINE
DATA AGGREGATION
USER
Local User
AI / ML
Anomaly detection
Generation / AQI Forecast
Peak demand Prediction
Environmental/Zone Comparison
RECOMMENDATION ENGINE
Alerts
What if
Optimize Route/Frequency
Trigger Air Filtration
Dynamic pickup schedule
GRAPH, REPORT, DASHBOARD
Graph
Reports
Waste Diverted
AQI Trends
Sustainable Impact
Dashboard
Admin
Client
Officer

BUSINESS MODEL

Smart. Sustainable. Scalable.

An AI-powered Facility Intelligence Platform for Modern Institutions

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STEP 1 - ONE-TIME

Hardware & Installation

Sensors, deployment & setup

One-Time Hardware Cost (Approx.)

₹1,00,000 - ₹1,50,000

Our team inspects the site, and the exact cost is shared after the analysis.

EstateIQ Basic

₹3,499 / month

₹41,988 / year

- All Four Facility Dashboards
- Basic ELITE Monitoring
- Monthly Reports & Alerts

EstateIQ Professional

₹5,999 / month

₹71,988 / year

- Advanced ELITE Analytics
- AI-Powered Decision Support
- Forecasting & Insights

EstateIQ Capabilities	EstateIQ Basic	EstateIQ Professional
SHAP Explainability & Transparent Insights	✓	✓
What-if Scenario Simulation	-	✓
Post-Action Verification	-	✓
AI Decision Support & Recommendations	✓	✓
Predictive Analytics & Forecasting	✓	✓
Offline Deployment Support	-	✓
Advanced Reports & Insights	Monthly	Up to 10× Advanced

Flexible Revenue Options

- **Hardware & Installation (One-Time)** — Sensors, deployment & setup
- **Subscription Plans (Monthly / Yearly)** — Basic to Pro
- **Self-Hosted License** — (On-Premise Deployment)
- **Source Code Access** — (with License)
- **Custom Solutions** — (For Large Campuses)

Why We're Different?

- **AI-Powered Insights** — Smarter decisions, not just data
- **Four-in-One Monitoring** — Energy + Water + Air + Waste in one platform
- **Cost-Effective & Scalable** — Built for Tier 2-3 Institutions
- **Trusted & Future-Ready** — Sustainable campuses, better tomorrow

FUTURE SCOPE

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From decision support to a trusted, ecalable estate platform

TODAY

EstateIQ monitors, predicts, explains and recommends. The admin decides.

TOMORROW

EstateIQ pinpoints faulty, energy-wasting equipment, recommends efficient replacements, and verifies savings after upgrades—with admin approval.

SHORT TERM

Validate

- Pilot in a real campus
- Human-in-the-loop trials
- Benchmark accuracy against industry standards

MID TERM

Expand

- Multi-building, cross-zone coupling
- Air quality, waste and mobility modules
- Partner hardware rollout, automated ESG reports

LONG TERM

Scale

- Portfolio view across campuses and cities
- Privacy preserving learning across sites

MORE RESOURCES

Air quality | Waste management | Mobility and fleets | Carbon and renewables

MORE SECTORS

Healthcare | Education | Industry | Smart cities | Government

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CONCLUSION

Facilities don't lack data. They lack decisions.

THE PROBLEM

Fragmented data

- Separate meters
- Manual analysis
- Late decisions

OUR ANSWER

Decisions, explained

- Predict and explain
- Tests the fix first
- Verifies the result

THE EDGE

Affordable, flexible

- On-site or cloud
- Pay for AI when needed
- Grows to all and waste

Monitor. Explain. Decide. Before the bill arrives.

Thank you • Team Elite • Questions welcome